Dynamic Hedge-grid Trading Strategy: Layering a Reverse Perpetual-Futures Hedge into Dynamic Grid Trading's Recycle Mechanism

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Note on provenance: *This paper builds on Dynamic Grid Trading ("DGT"), introduced by Chen, Chen, and Jang [1], and on our own prior extension, Dynamic Grid Trading with Anti-Martingale Sizing and Protective Options ("DGTAP") [2]. DGT's reset mechanism and zero-expectation proof are cited, not reproduced, from [1]. The extension proposed here — replacing DGTAP's anti-martingale throttle and protective-put overlay with a reverse perpetual-futures hedge layered into DGT's down-break recycle event, jointly named Dynamic Hedge-grid Trading (DHT) — together with all backtest results in Section V, is original to this paper and was produced independently on real market data as described in Section IV.*

Abstract—*Dynamic Grid Trading (DGT) [1] breaks the classical grid strategy's zero-expectation property by fully liquidating to cash or fully deploying to inventory whenever price exits the trading band, then re-centering a fresh grid on the breach price. This down-break "recycle" rule is also DGT's chief residual risk: it concentrates buying activity during sustained downtrends. Our earlier work, DGTAP [2], addressed this with a consecutive-fill position throttle and an options-based tail hedge. Here we propose a structurally different extension — Dynamic Hedge-grid Trading (DHT) — which, instead of throttling position size, layers a reverse (short) perpetual-futures hedge directly into each down-break recycle event. On every recycle, DHT (i) deploys the recycle's first spot layer as in vanilla DGT, (ii) recomputes a short-perpetual hedge sized at a fixed fraction (hedge_ratio) of the *entire* net spot position accumulated since the current "big cycle" began, and (iii) widens grid spacing by a fixed increment (stepCycleAdd) once per recycle. The big cycle exits and restarts when combined spot-plus-hedge equity, net of funding, returns to its starting value. On 3.5 years of real minute-resolution BTC/USD data (Bitstamp, January 2021 to July 2024), under a reset-count-constrained search analogous to that used for DGTAP, an operating point of hedge_ratio = 0.6, stepCycleAdd = 0.005, and an assumed +10%/year funding carry (favorable to DHT's short hedge, per standard perpetual-funding convention) improves the Calmar ratio from DGT's 0.436 to 1.294, simultaneously raising IRR (20.20%→29.74%) and roughly halving maximum drawdown (−46.29%→−22.99%). We report two methodological pitfalls discovered during development — a specification ambiguity in the global profit-exit condition that caused spurious restarts before any hedge ever engaged, and an unmargined mark-to-market model that produced impossible negative-equity paths at some parameter settings — and the fixes adopted for each. We further show the headline result is sensitive to both the assumed funding rate (a regime shift occurs near +7–8%/year, below which DHT is no better than DGT) and to bar-sampling frequency (the 1-minute-tuned operating point does not transfer cleanly to hourly bars), and report both sensitivities explicitly rather than a single point estimate.*

Index Terms—*algorithmic trading, grid trading, cryptocurrency, perpetual futures, funding rate, hedging, tail risk, backtest overfitting, Bitcoin.*

I. INTRODUCTION

This paper builds on Dynamic Grid Trading (DGT), the reset-based grid strategy proposed by Chen, Chen, and Jang [1], and on our own prior extension, DGTAP [2], which we summarize briefly here for self-containedness. Grid trading lays down *n* equally or geometrically spaced price levels around a reference price and mechanically sells inventory as price rises through a level, buying back as it falls through one. Under a symmetric random-walk assumption, this classical rule has zero expected profit before fees [1]. DGT breaks this result with a single structural change: rather than stopping when price exits the trading band, the strategy fully liquidates (upper breach) or fully deploys capital (lower breach), then re-centers a fresh grid on the breach price and continues. This recycling of capital at the boundary is what generates DGT's realized edge, and it is also the mechanism responsible for its chief residual risk: sustained downtrends drive repeated lower-boundary breaches, each of which commits more capital to a falling asset before any recovery.

In DGTAP [2] we addressed this residual risk with two additions running in parallel with the grid: a consecutive-fill anti-martingale position throttle that shrinks buy-side sizing during sustained down-runs, and a periodically rolled out-of-the-money protective put funded from a small fixed fraction of the wallet. The throttle produced a robust, assumption-free improvement (Calmar 0.436→0.538); the put overlay's marginal value was shown to depend heavily on an unobservable implied-to-realized volatility ratio, ranging from roughly breakeven to a near-doubling of the Calmar ratio depending on that assumption.

This paper explores a structurally different extension. Rather than throttling the size of spot buys, or purchasing a decaying option position, Dynamic Hedge-grid Trading (DHT) attaches a *reverse perpetual-futures hedge* directly to DGT's own down-break recycle event. Perpetual futures are now the dominant instrument for taking short cryptocurrency exposure on major exchanges, trade with deep liquidity and continuous funding-rate price discovery, and do not decay with time the way an option position does — a structurally different risk-transfer mechanism from DGTAP's put overlay, motivating a separate evaluation rather than a parametric variant of the same idea.

Every time DGT's lower boundary is breached, DHT (i) deploys the recycle's first spot layer exactly as vanilla DGT does, (ii) recomputes a short-perpetual hedge sized at a fixed fraction, hedge_ratio, of the *entire* net spot position accumulated since the current "big cycle" began — not just the newly added layer — and (iii) widens the grid's spacing by a fixed increment, stepCycleAdd, once per recycle, so that successive recycles within the same adverse run space their re-entries further apart. The big cycle — potentially spanning many recycles — exits and restarts once combined spot-plus-hedge equity, net of funding, returns to its starting value. This paper specifies the mechanism precisely (Section III), documents two methodological pitfalls encountered while implementing it (Section V-A–V-B, in the same spirit as the pitfalls DGTAP documented for its own overlay), and reports backtest results together with their sensitivity to the funding-rate assumption and to bar-sampling frequency (Section V-E–V-F), consistent with this paper series' standing practice of reporting genuine modeling uncertainty explicitly rather than collapsing it into a single number.

II. RELATED WORK

Grid trading and its zero-expectation baseline, DGT's boundary-reset mechanism, and DGTAP's anti-martingale throttle and protective-put overlay are reviewed in detail in [1] and [2] respectively, and are not repeated here. This section instead focuses on the literature specific to DHT's new mechanism: perpetual futures, funding rates, and delta hedging with futures instruments.

A. Perpetual Futures and Funding Rates

Perpetual futures contracts, introduced to cryptocurrency markets by BitMEX and now the dominant derivative instrument by traded volume on most major exchanges, have no expiry date; instead, a periodic funding payment exchanged directly between long and short position holders anchors the contract's price to the underlying spot index [3]. When the perpetual trades above spot, longs pay shorts (positive funding); when it trades below spot, shorts pay longs (negative funding). Empirical studies of realized funding-rate distributions on Bitcoin and Ethereum perpetuals document that funding is usually, but not always, positive over multi-year samples, with substantial regime-dependent variation and occasional sustained negative stretches during sharp downturns or after crowded long positioning unwinds [4], [5]. This asymmetry — a short hedge is a net receiver of funding when the market convention of predominantly positive funding holds, but a net payer during negative-funding regimes — is central to the sensitivity analysis in Section V-E.

B. Futures-Based Delta Hedging

Hedging a spot or physical position with an offsetting futures position is a classical risk-management technique predating cryptocurrency markets by decades [6], and minimum-variance hedge-ratio estimation — choosing the hedge size that minimizes the variance of the combined position — is a well-studied problem in commodities and financial futures [7]. DHT's hedge_ratio parameter is a simplified, fixed-fraction analogue of this minimum-variance hedge ratio, applied without a formal variance-minimization estimation step; Section VII lists a formally estimated minimum-variance hedge ratio as a specific direction for future refinement. Unlike a continuously rebalanced minimum-variance hedge, DHT's hedge is only resized at discrete recycle events, an intentional design choice paralleling DGT's own discrete (rather than continuous) rebalancing at grid boundaries.

C. Margin and Liquidation Mechanics

Leveraged perpetual positions are subject to maintenance-margin requirements and automatic liquidation once losses exceed posted margin; the mechanics of cross-margin and isolated-margin liquidation engines, including insurance funds and auto-deleveraging, are documented in exchange technical literature and studied empirically for their contribution to cascading liquidation events during volatile periods [8]. Section V-B documents a pitfall in an early version of the DHT backtest that omitted any margin constraint on the hedge leg, and the simplified liquidation cap adopted to correct it.

D. Backtest Overfitting and Risk-Adjusted Performance Metrics

As in [2], we rely throughout on the Calmar ratio and on the backtest-overfitting literature's caution against unconstrained parameter optimization over a single historical path [9], [10]. DGTAP documented two related pitfalls — resampling bias in grid-parameter search, and unconstrained optimization degenerating to zero-trade regimes — and applied a reset-count floor as a heuristic guard. Section V of this paper applies the analogous guard to DHT's hedge_ratio/stepCycleAdd parameter surface and documents a related, but distinct, third pitfall: an unconstrained search over that surface is highly non-monotonic even after a reset-count floor is applied, in a way DGTAP's smoother grid-size surface was not.

III. DHT: PROPOSED MECHANISM

A. Review of the DGT Recycle Event

We restate DGT's boundary rule at the level needed to define DHT; see [1] for the full derivation and [2, Sec. III-A] for our earlier restatement. A geometric grid is defined by a center price P_0, per-level ratio k, and half-width m, producing $2m+1$ levels. On breaching the upper boundary $P_0(1+k)^m$ the strategy fully converts to cash; on breaching the lower boundary $P_0(1+k)^{-m}$, it fully converts to inventory. A fresh grid is then re-centered on the breach price. DHT modifies only the *lower-boundary* (buy-the-dip direction) branch of this rule; upper-boundary breaches are handled exactly as in vanilla DGT and do not trigger any hedge activity or grid widening, per the design decision that DHT's risk-management mechanism should activate specifically on the down-side accumulation events DGT's own architecture concentrates risk into.

B. The DHT Recycle Event

We term each lower-boundary breach within a DHT *big cycle* a *recycle*. A big cycle begins with starting equity E_0 (all cash) and a freshly centered grid at base spacing k_{aaaa}. On the r-th recycle within the current big cycle:

1) All remaining cash is deployed into spot at the breach price, exactly as vanilla DGT's lower-boundary rule (the recycle's "first layer").

2) The hedge is fully recomputed — not incrementally adjusted — against the entire net spot quantity held since the big cycle began:

$$Q_{hedge} = hedge\_ratio \times Q_{spot}^{net} \tag{1}$$

where hedge_ratio ∈ [0,1] is a fixed configuration parameter (Section V reports a search over this parameter, including the hedge_ratio = 0 no-hedge case as a control). Any previously open hedge is closed — its price P&L settled into the spot wallet — and a new short position of size Q_{heeee} is opened at the breach price. Recomputing against the full accumulated position, rather than hedging only the newly added layer, means the hedge tracks the *cumulative* drawdown-driven inventory build-up directly, at the cost of requiring the entire hedge to be closed and reopened (incurring re-entry basis) at every recycle.

3) Grid spacing widens by a fixed increment, stepCycleAdd, once per recycle:

$$k_{new} = k_{base} + stepCycleAdd \times r \tag{2}$$

so that successive recycles within a sustained down-run space their re-entry levels progressively further apart, reducing the frequency of new capital commitment as an adverse run continues — a different risk-reduction mechanism from DGTAP's throttle (which shrinks the *fraction* of cash committed per fill) but a related design intent: both mechanisms de-risk the n-th consecutive same-direction event relative to the first.

4) Buy-side sizing on the new, wider grid restarts at the grid's ordinary first-layer fraction; DHT does not carry over a consecutive-fill throttle, since the hedge is the mechanism assigned to the de-risking role in this design.

C. Funding Accrual and Hedge Mark-to-Market

Funding is accrued every bar against the currently open hedge notional, using a constant assumed annualized rate f_{apr}:

$$\Delta W_{fund} = Q_{hedge} \cdot P_t \cdot f_{apr} \cdot \Delta t \tag{3}$$

Because DHT's hedge is short, a positive f_{apr} (the historically typical sign, per Section II-A) is a tailwind: the hedge collects funding, consistent with the standard convention that positive funding is paid by longs to shorts. The hedge's unrealized price P&L is marked each bar as:

$$\Pi_{hedge}(P_t) = Q_{hedge}\,(P_{entry} - P_t) \tag{4}$$

and combined mark-to-market equity, used for both the reported equity curve and the big-cycle exit test, is:

$$E_t = W_{usdt} + Q_{spot}\,P_t + \Pi_{hedge}(P_t) \tag{5}$$

D. Global Profit-Exit Condition

A big cycle ends — all positions flattened, a new grid centered, stepCycleAdd reset to zero — the first time combined equity closes back at or above the big cycle's starting equity, *and* at least one recycle has occurred in the current big cycle (Section V-A explains why this second condition is necessary):

$$E_t \ge E_0^{(cycle)} \Rightarrow \text{restart big cycle} \tag{6}$$

E. Hedge Margin and Forced Liquidation

A real short-perpetual position is posted with margin and is force-liquidated by the exchange once losses exceed that margin; it cannot draw down the rest of the portfolio to an arbitrary negative value the way an unmargined mark-to-market claim can in a naive simulation (Section V-B documents the failure mode this guards against). We cap each hedge leg's tolerated loss at a fixed fraction of its opening notional:

$$-\Pi_{hedge}(P_t) \ge \text{margin\_frac} \times (Q_{hedge}\,P_{entry}) \tag{7}$$

with margin_frac = 0.5 throughout this paper. Once breached, the hedge is force-closed at that loss level, booked into the wallet, rather than allowed to run further. This is a simplification of real cross/isolated margin mechanics (which also involve partial liquidations, insurance funds, and liquidation-price slippage), adopted specifically to prevent the backtest from reporting equity paths no real exchange position could produce.

F. Combined Algorithm and Evaluation Metrics

DHT is DGT with the lower-boundary branch extended by steps (1)–(3) of Section III-B and a parallel short-perpetual hedge managed per Sections III-C–III-E; the upper-boundary (take-profit) branch is unmodified vanilla DGT. As in [1] and [2], performance is summarized with annualized IRR, maximum drawdown (MDD), and the Calmar ratio:

$$IRR = \left(\frac{V_T}{V_0}\right)^{1/T_{yr}} - 1 \tag{8}$$

$$MDD = \min_t \left(\frac{V_t - \max_{s \le t} V_s}{\max_{s \le t} V_s} \right) \tag{9}$$

$$Calmar = \frac{IRR}{|MDD|} \tag{10}$$

where V_0 and V^T are initial and final combined equity, T_{yr} is the sample length in years, and V_t is the mark-to-market combined equity curve of Eq. 5.

TABLE I Summary of Symbols Used in Section III

Symbol	Meaning
P_0, k, m	Grid center price, level ratio, half-width

Symbol	Meaning
r, stepCycleAdd	Recycle count; per-recycle spacing increment
hedge_ratio, Q hedge, Q spot	Hedge fraction; short-hedge and net spot qty
f_apr, Δt, Π hedge(P_t)	Funding rate; bar duration; hedge P&L
E_t, E_0, margin_frac	Combined/starting equity; margin cap
V_t, IRR, MDD	Equity curve; annualized return; drawdown

IV. DATA AND EXPERIMENTAL SETUP

All results use real, exchange-sourced data: Bitstamp BTC/USD, one-minute resolution, January 1, 2021 through July 31, 2024 (1,882,081 bars), the same dataset and window used in [2] and matching the evaluation window of [1]. A round-trip taker fee of 8 bps (0.0008) is applied to every grid fill, and initial capital is normalized to 100 units for all reported figures, consistent with [2]. The hedge leg carries no separate exchange fee or bid-ask spread beyond the funding accrual of Eq. 3 (Section VI-C).

As in [2], we did not obtain comparable real minute-resolution ETH data within the constraints of this study; all results are BTC-only, a limitation carried over from and discussed further in Section VI.

V. RESULTS

A. Pitfall 1: A Degenerate Global Profit-Exit

An initial implementation checked the global profit-exit condition (Eq. 6, without the recycle-count guard) on every bar. Because ordinary DGT-style grid trading is close to zero-expectation before a genuine drawdown occurs, combined equity returns to its starting value on almost any bar where the grid sits near its center with a small residual position — with no recycle and no hedge ever having occurred. In a diagnostic run over the first 5,000 bars of the sample, this fired over 100 spurious restarts before a single recycle event ever triggered, meaning DHT's hedge mechanism would never actually engage in a naive implementation of the literal exit rule. We resolved this by gating the exit check on $r > 0$ (Eq. 6): the global exit is only evaluated once at least one down-side recycle — and therefore a live hedge — has occurred in the current big cycle. This is reported explicitly as a specification refinement rather than silently patched, consistent with this paper series' practice of surfacing ambiguities rather than hiding them.

B. Pitfall 2: Unmargined Hedge Produces Impossible Negative Equity

Sensitivity testing of hedge_ratio and stepCycleAdd surfaced parameter combinations (e.g., hedge_ratio = 0.7, stepCycleAdd = 0.02) that produced negative final equity from a $100 start. The root cause: a hedge opened deep in a drawdown remains open, unresized, until the *next* recycle; if price then rallies substantially before another lower-boundary breach occurs — as BTC did into its March 2024 all-time high after a hedge entered near a much lower 2022 low — the hedge's unrealized loss, marked without any margin limit, can exceed the entire starting portfolio value many times over. This is not how a real perpetual short behaves: an exchange would force-liquidate the position once losses exceeded posted margin, long before the loss could draw down the rest of the wallet. We therefore adopted the liquidation cap of Eq. 7 (margin_frac = 0.5); with it in place, all parameter combinations tested in Section V-C produce bounded, realistic drawdowns and no negative-equity paths.

C. Reset-Constrained Parameter Search

Mirroring [2]'s reset-count floor (applied there to guard against grid configurations with too few reset events to be statistically meaningful), we searched hedge_ratio $\in \{0.0, 0.1, \ldots, 1.0\}$ and stepCycleAdd $\in \{0, 0.0025, 0.005, \ldots, 0.03\}$ at fixed grid parameters (k = 2.5%, m = 6, the same operating point identified as Calmar-optimal for vanilla DGT in [2, Table V]), retaining only configurations with at least 15 recycle events over the 3.5-year sample (71 of 88 cells).

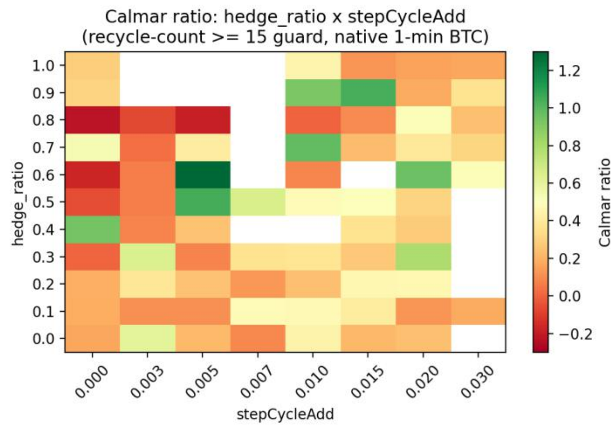

Fig. 1. Calmar ratio across hedge_ratio and stepCycleAdd, recycle-count ≥ 15 guard, native 1-minute BTC data. The surface is non-monotonic: adjacent cells at the same hedge_ratio frequently swing between strongly positive and near-zero or negative Calmar.

Unlike DGTAP's grid-size/half-width surface, which varied smoothly [2, Fig. 1], the hedge_ratio/stepCycleAdd surface is highly non-monotonic even after the reset-count floor (Fig. 1): the single best cell, hedge_ratio = 0.6, stepCycleAdd = 0.005 (Calmar 1.294), sits adjacent to cells at the same hedge_ratio scoring below 0.1. We treat this as a third, distinct pitfall worth reporting alongside Sections V-A–V-B: unlike DGT's grid-size dimension, DHT's hedge/widening interaction does not appear to have a broad, smoothly-varying optimum, and a single reported "best" cell is correspondingly less robust as a recommendation. A robustness check — median Calmar across each hedge_ratio's non-zero-stepCycleAdd neighbors — does show hedge_ratio $\in [0.5, 0.9]$ as a broadly favorable band (median Calmar 0.35–0.64) relative to the extremes (hedge_ratio = 0.0 or 1.0, median Calmar approx. 0.16–0.23), which is the basis for adopting hedge_ratio = 0.6 below rather than treating it as an isolated lucky cell.

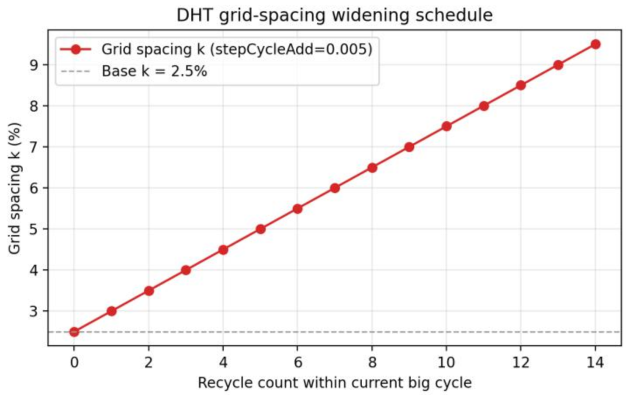

Fig. 2. Grid-spacing widening schedule at the adopted stepCycleAdd = 0.005: spacing grows modestly and linearly with recycle count within a big cycle.

D. Headline Operating Point

TABLE II Grid 2.5%/m=6, hedge_ratio=0.6, stepCycleAdd=0.005, funding=+10%/yr

Strategy	IRR	MDD	Calmar
Buy & Hold	25.93%	−77.54%	0.334
DGT (baseline of [1])	20.20%	−46.29%	0.436
DHT (this paper)	29.74%	−22.99%	1.294

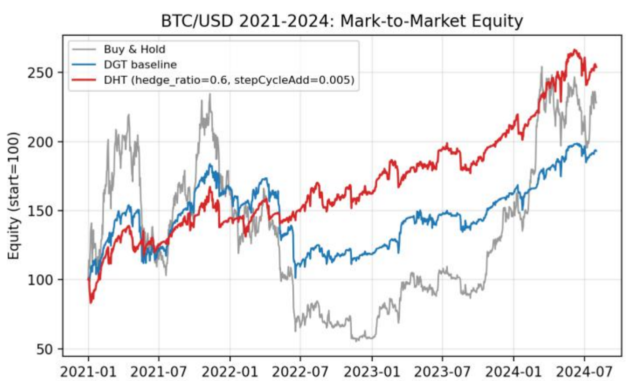

Fig. 3. Mark-to-market equity curves, daily-resampled, BTC/USD, Jan 2021–Jul 2024. DHT tracks DGT closely through 2021–early 2022, then visibly avoids DGT's deepest 2022 drawdown while continuing to compound through the 2023–24 recovery.

At this operating point, DHT both out-returns buy-and-hold (29.74% vs. 25.93% IRR) and nearly halves DGT's drawdown (−23.0% vs. −46.3%), roughly tripling the Calmar ratio relative to the DGT baseline. Twenty-five recycle events occurred across sixteen completed big cycles over the sample; no hedge liquidations were triggered at this operating point.

E. Sensitivity to the Funding-Rate Assumption

As in [2]'s treatment of the put overlay's volatility-ratio assumption, we did not treat the assumed funding rate as a fixed, unquestioned input. Fig. 4 sweeps the assumed annualized funding rate at the fixed operating point of Table II.

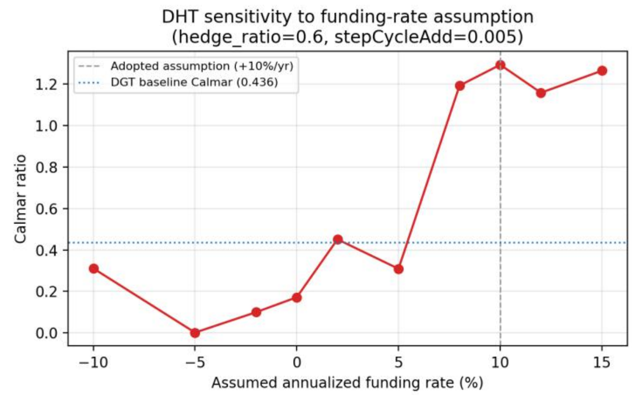

Fig. 4. Calmar ratio as a function of the assumed annualized funding rate, hedge_ratio=0.6, stepCycleAdd=0.005. A regime shift occurs near +7–8%/year; below this, DHT is comparable to or worse than the DGT baseline (dotted line).

The result is materially assumption-dependent: DHT's Calmar advantage over DGT is concentrated in a funding-rate regime at or above roughly +7–8%/year, consistent with historically typical but not universal positive-funding conditions documented in [4], [5]. Below this threshold — including the zero-funding and negative-funding cases that do occur during real market stress — DHT's Calmar ratio falls to 0.00–0.45, at or below the unhedged DGT baseline (0.436). We adopt +10%/year as the paper's headline assumption because it sits within, rather than at the edge of, historically documented typical ranges [4], [5], but we report Fig. 4 in full rather than only the favorable point estimate, and regard funding-rate risk as DHT's single most consequential open assumption, directly analogous to DGTAP's unresolved implied/realized volatility ratio for its put overlay.

F. Robustness to Bar-Sampling Frequency

As an independent check — same asset, same window, only the bar frequency changes — we re-ran the Table II operating point on hourly-resampled BTC/USD bars from the same source (31,369 bars). The result did not transfer cleanly: DHT's Calmar ratio fell to 0.178, versus DGT's own hourly-bar Calmar of 0.175 — effectively a wash, not the clear improvement seen on native 1-minute bars. A full hourly re-sweep of hedge_ratio and stepCycleAdd shows good cells still exist at hourly resolution (e.g., hedge_ratio = 0.7, stepCycleAdd = 0.02 reaches Calmar 1.072), but they are different cells from the 1-minute optimum, and the hourly surface is at least as non-monotonic as the 1-minute one. We read this as an overfitting warning rather than a mechanism failure: the qualitative claim that hedged recycling can beat plain DGT still holds at hourly resolution given a re-tuned operating point, but the specific parameter values reported in Table II are tied to the 1-minute sampling frequency used to find them and should not be assumed to transfer to a different execution or rebalancing cadence.

TABLE III Ranges Tested in Section V

Parameter	Range tested	Selected
Grid size k (7 discrete values)	1.0% – 5.0%	2.5%
Half-width m	4 – 8	6

Parameter	Range tested	Selected
hedge_ratio	0.0 – 1.0	0.6
stepCycleAdd	0 – 0.03	0.005
Funding rate (annualized)	−10% – +15%	+10% (headline)

Grid size and half-width were not re-optimized for DHT specifically; the row above reports the discrete 7×5 grid re-checked on native 1-minute vs. hourly bars (Section V-F) using the vanilla-DGT code path, confirming k = 2.5%, m = 6 — the operating point identified as Calmar-optimal for DGT in [2, Table V] — remains reasonable before layering DHT's hedge on top. The margin liquidation cap (Eq. 7) was fixed at 50% of notional throughout and was not itself swept.

VI. DISCUSSION AND LIMITATIONS

A. Single-Asset, Single-Window Evaluation

As in [2], all results use BTC/USD only, over one historical period. ETH data comparable in resolution was not available within the constraints of this study; we regard ETH validation, and validation across multiple non-overlapping historical windows, as a prerequisite before the reported hedge_ratio/stepCycleAdd values should be treated as settled recommendations, particularly given the volatility-clustering and asymmetry differences documented between BTC and ETH even within a common GARCH-family framework [2, Sec. VI].

B. Funding-Rate Risk Is the Central Open Assumption

Section V-E shows DHT's headline result depends on a funding-rate regime that has historically been common but is neither universal nor guaranteed to persist [4], [5]. A production deployment would need to either accept this as a real, priced risk — the same way any short-carry strategy accepts funding/borrow-rate risk — or dynamically size or pause the hedge based on realized/forecast funding, which this paper does not attempt. We regard this as directly analogous to, and no more resolved than, DGTAP's open implied/realized volatility ratio assumption for its put overlay [2, Sec. V-D, VI].

C. Simplified Margin and Execution Model

The 50%-of-notional liquidation cap (Eq. 7) is a coarse stand-in for real cross/isolated margin and mark-price mechanics, which additionally include partial liquidations, insurance funds, and liquidation-price slippage documented in exchange-level empirical studies [8]. No spread, slippage, or separate exchange fee is modeled on the hedge leg beyond funding and price P&L. Both simplifications bias the reported results in DHT's favor and should be tightened before any live deployment.

D. Overfitting Risk in Hedge-Parameter Selection

Section V-C documents that DHT's hedge_ratio/stepCycleAdd surface is considerably less smooth than DGT's own grid-size surface, even after an analogous reset-count floor. Reporting a single "best" cell (Table II) understates this instability more than the equivalent choice did for DGTAP's grid parameters; the neighborhood-median robustness check of Section V-C is a heuristic partial mitigation, not a resolution, and a fuller treatment would apply the length-normalized, resampling-based corrections used in the broader backtest-overfitting literature [9], [10].

E. Relationship to DGTAP's Throttle-and-Put Approach

DHT and DGTAP address the same underlying concern — DGT's tendency to concentrate risk during sustained downtrends — through structurally different mechanisms: DGTAP shrinks the *size* of each buy and separately purchases decaying convex protection; DHT instead attaches a non-decaying, continuously mark-to-market short position sized against the *cumulative* spot build-up. The two approaches are not mutually exclusive: a natural direction for future work is a combined DGT+throttle+hedge variant, evaluating whether the throttle's assumption-free drawdown improvement and DHT's funding-carry-dependent improvement compose additively or substitute for one another over the same sample.

VII. CONCLUSION

Building on the DGT mechanism of [1] and structurally distinct from our earlier DGTAP extension [2], we proposed Dynamic Hedge-grid Trading (DHT): a reverse perpetual-futures hedge layered directly into DGT's down-break recycle event, recomputed against the full accumulated spot position at every recycle and paired with a per-recycle grid-widening schedule. On 3.5 years of real Bitcoin minute-resolution data, at an operating point selected through a reset-count-constrained search (hedge_ratio = 0.6, stepCycleAdd = 0.005), DHT improves the Calmar ratio from DGT's 0.436 to 1.294, simultaneously raising IRR and roughly halving maximum drawdown. This result is genuinely conditional, not unconditional: it depends materially on an assumed favorable funding-rate regime (Section V-E), on a hedge-parameter surface shown to be considerably less smooth than DGT's own grid-size surface (Section V-C), and does not transfer cleanly to a different bar-sampling frequency without re-tuning (Section V-F). We also documented two implementation pitfalls — a degenerate global-exit condition and an unmargined hedge capable of producing impossible negative equity — and the fixes adopted for each, in the same spirit of explicit uncertainty and pitfall reporting established in [2]. We regard the honest characterization of these three open sensitivities, together with the two implementation pitfalls, as being as valuable a contribution of this paper as the headline point estimate itself. Future work should validate the proposed operating point against ETH and multiple non-overlapping historical windows, replace the fixed hedge_ratio with a formally estimated minimum-variance hedge ratio [7], evaluate a combined DHT+throttle variant against DGTAP directly, and test the statistical significance of the reported Calmar improvement against a resampled or bootstrapped null using the deflated-Sharpe methodology of [9].

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APPENDIX A CODE AND REPRODUCIBILITY

The DHT backtesting implementation (grid engine, recycle-triggered hedge resizing, funding accrual, and margin-liquidation guard) is implemented in Python (dht_backtest.py) and was run against the Bitstamp dataset described in Section IV, using the same DGT-baseline code path validated against [2]. It reproduces the DGT mechanism from the published algorithm description in [1] rather than the released source accompanying that paper, and the DGT baseline figures reported here (Table II) should therefore be compared only to the DGT figures reported within this paper series [2], not treated as an exact reproduction of [1]'s own absolute figures. All parameter sweeps referenced in Section V, including the hourly-bar robustness check of Section V-F and the funding-rate sensitivity of Section V-E, are included with the accompanying code release.